1 Big Ideas

1.1 Theorem 8.6.1 (Existence of Real Numbers):

> There exists an ordered field in which every nonempty set that is bounded above has a least upper bound. In addition, this field contains $\mathbf{Q}$ as a subfield.

This chapter takes an axiomatic approach attributed to Dedekind Cuts. The Real Numbers ($\mathbf{R}$) is the set of all Dedekind Cuts. We will construct $\mathbf{R}$ in the following steps:

1. Define the Dedekind Cuts
2. Define an Order on the cuts
3. Define the Algebraic properties
4. Define Least Upper Bounds on the Cuts

2 Dedekind Cuts

2.1 Definition 8.6.2:

A subset A of the rational numbers is called a *cut* if it possesses the following three properties:

1. $A \neq \emptyset$ and $A \neq \mathbb{Q}$.
2. If $r \in A$, then A also contains every rational $q < r$.
3. A does not have a maximum; that is, if $r \in A$, then there exists $s \in A$ with $r < s$.

2.2 Exercise 8.6.1:

Problem 1.

(a) Fix $r \in \mathbf{Q}$. Show that the set $C_r = \{t \in \mathbf{Q} : t < r\}$ is a cut.

Solution. Fix $r \in \mathbf{Q}$ and let there exists the set $C_r = \{t \in \mathbf{Q} : t < r\}$. Given C_r $r-1$ but not r, we know both that $C_r \neq \emptyset$ and $C_r \neq \mathbf{Q}$, implying Definition 8.6.2 (1) is satisfied. Then, if $p \in C_r$ and $q < p$, then $q < p < r$, which implies that $q < r$ and $q \in C_r$, satisfying Definition 8.6.2 (2). Finally, we can find $t < \frac{t+r}{2} < r$ in C_r such that 8.6.2 (3) is satisfied, thus C_r is a cut. □

The temptation to think of all cuts as being of this form should be *avoided*. Which of the following subsets of $\mathbf{Q}$ are cuts?

(b) $S = \{t \in \mathbf{Q} : t \leq 2\}$

Solution. Given a set $S = \{t \in \mathbf{Q} : t \leq 2\}$, by Definition 8.6.2 (3), in order for S to be a cut, S cannot attain a maximum. However, S the maximum $s = 2$. So, S is not a cut. □

(c) $T = \{t \in \mathbf{Q} : t^2 < 2 \text{ or } t < 0\}$

Solution. Given a set $T = \{t \in \mathbf{Q} : t^2 < 2 \text{ or } t < 0\}$, take $-1 = t$, this satisfies the definition of T, so T is nonempty. Suppose $t = 2$, now none of the parts of T is satisfied, so $2 \notin T$, then $T \neq \mathbf{Q}$. Definition 8.6.2 (1) has been satisfied. Next, suppose $t \in T$ where $s < t$. The first case is where $s < 0$, in which by definition of T, $s \in T$. In the case where $s \geq 0$, it is implied that $0 \leq s < t$. We must have that $t^2 < 2$ here since $t \in T$. So,

$$s^2 < t^2 < 2 \implies s^2 < 2$$

, and thus $s \in T$. Definition 8.6.2 (2) is satisfied. Let $t \in T$ where $s > t$. In the case where $t < 0$, set $s = 0$. Then, $s^2 < 2$, and thus $s \in T$. It $t^2 \geq 0$, set then it must be true that $t^2 < 2$, meaning $t < \sqrt{2}$. Since $\mathbf{Q}$ is dense in $\mathbf{R}$, there exists a s such that $t < s < \sqrt{2}$. Thus, Definition 8.6.2 (3) is satisfied. So, T is a cut! □

(d) $U = \{t \in \mathbf{Q} : t^2 \leq 2 \text{ or } t < 0\}$

Solution. Given a set $U + \{t \in \mathbf{Q} : t^2 \leq 2 \text{ or } t < 0\}$. Let $t = -1$, since $-1 < 0$, $t \in U$, and is thus nonempty. Set $t = 2$, then neither condition of the defintion of U is fulfilled, and thus $U \neq \mathbf{Q}$, satisfying Definition 8.6.2 (1). If there exists an $s \in U$, there is two cases. One where $s < 0$, meaning it is bounded above. Another where $s \geq 0$. This case implies that since $s < t$, and $t > 0$. This means $s \geq 0$ must be an element where $t^2 \leq 2$ since $t \not< 0$. Now

$$0 \leq s < t \implies s^2 < t^2 \leq 2 \implies s^2 < 2,$$

thus in U, satisfying Definition 8.6.2 (2). Finally, when $t \in U$, let $s \in U$ such that $s > t$. Then, two cases emerge, if $t < 0$, set $s = 0$. Then, $s^2 \leq 2$, and thus $s \in U$. If $t \geq 0$, then $t^2 \leq 2$. Since $\sqrt{2}$ is irrational, we can assume this implies that $t^2 < 2$, meaning that $t < \sqrt{2}$. By the density of $\mathbf{Q}$ in $\mathbf{R}$, there exists an s such that $t < s < \sqrt{2}$. Thus, U obtains no maximum and Definition 8.6.2 (3) is fulfilled. So, U is a cut! □

2.3 Exercise 8.6.2:

Problem 2. Let A be a cut. Show that if $r \in A$ and $s \notin A$, then $r < s$.

Solution. Let A be a cut and $r \in A$ but $s \notin A$. Assume for contradiction purposes that $r \geq s$. Since A is a cut, by Definition 8.6.2 (2), A must contain every rational $q < r$. This directly contradicts the assumption that $r \geq s$. So, it must be true that $r < s$. □

2.4 Definition 8.6.3:

Define the *real numbers* $\mathbb{R}$ to be the set of all cuts in $\mathbb{Q}$.

3 Field and Order Properties

3.1 Definition 8.6.4:

A set F is a *field* if there exists two operations- addition $(x + y)$ and multiplication (xy)- that satisfy the following list of conditions:

1. (commutativity) $x + y = y + x$ and $xy = yx$ for all $x, y \in F$.
2. (associativity) $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$ for all $x, y, z \in F$.
3. (identities exist) There exist two special elements 0 and 1 with $0 \neq 1$ such that $x+0 = x$ and $x1 = x$ for all $x \in F$.
4. (inverses exist) Given $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$. If $x \neq 0$, there exists an element x^{-1} such that $xx^{-1} = 1$.
5. (distributive property) $x(y + z) = xy + xz$ for all $x, y, z \in F$.

3.2 Definition 8.6.5:

An *ordering* on a set F is a relation, represented by $\leq$, with the following three properties:

1. For arbitrary $x, y \in F$, at least one of the statements $x \leq y$ or $y \leq x$ is true.
2. If $x \leq y$ and $y \leq x$, then $x = y$
3. If $x \leq y$ and $y \leq z$, then $x \leq z$

   A field F is called an *ordered field* if F is endowed with an ordering $\leq$ that satisfies

4. If $y \leq z$, then $x + y \leq x + z$.
5. If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$.

4 Where are we at?

Let's examine what we have done thus far. We have accepted that **Q** are an ordered field. Now, we must implement addition, multiplication, and an ordering that possesses these definitions. The easiest of the next steps is *ordering*. Let A and B be two arbitrary elements of **R**.

Define $A \leq B$ to mean $A \subseteq B$.

4.1 Exercise 8.6.4

Problem 3. Show that this defines an ordering on $\mathbf{R}$ by verifying properties of ordering from Definition 8.6.5.

Solution. To prove 8.6.5 (1), begin by assuming $A \not\subseteq B$. By definition, this means there is some $a \in A$ with $a \notin B$. From Exercise 8.6.2, this means that for all $b \in B$, $b < a$. Then by Definition 8.6.2 (2) $b \in A$; hence $B \subseteq A$. 8.6.5 (2) is proved by a direct result of the definition of set equality. Finally, 8.6.5 (3) is proved by transitivity of the set inclusion relationship. □

5 Algebra R

5.1 Definition:

Given $A, B \in \mathbf{R}$, define

$$A + B = \{a + b : a \in A \quad \text{and} \quad b \in B\}.$$

Let $a + b \in A + B$ be arbitrary and let $s \in \mathbf{Q}$ satisfy $s < a + b$. Then, $s - b > a$, which implies that $s - b \in A$ because A is a cut. But then

$$s = (s - b) + b \in A + B,$$

and Definition 8.6.2 part 2 is proved.

5.2 Exercise 8.6.5:

Problem 4.

(a) Show that Definition 8.6.2 part 1 and part 3 also hold for $A + B$. conclude that $A + B$ is a cut.

Solution. We just showed how $s \in A + B$. Thus, $A + B$ is nonempty. Now, let there be upper bounds where $l_1 \notin A$ and $l_2 \notin B$. For any $a \in A$, we have $l_1 > a$ and for any $b \in B$, we have $l_2 > b$. Thus, we have $(l_1 + l_2) > (a + b)$ for any $(a + b) \in A + B$. Since $\mathbf{Q}$ is unbound, $A + B \neq \mathbf{Q}$. Definition 8.6.2 (1) is satisfied.

Fix a $c \in A + B$ such that $c = a + b$. Given A and B are cuts, there exists $s \in A$ where $s > a$ and $r \in B$ with $r > b$. Now, $c = (a + b) < (s + r)$, implying no maximum. This satisfies Definition 8.6.2 (3)

Given all parts of Definition 8.6.2 are satisfied, $A + B$ cam be defined as a cut. □

(b) Check that addition in $\mathbf{R}$ is commutative (Defn 8.6.4 part one) and associative (Defn 8.6.4 part two).

Solution. First, let there exist an arbitrary $a+b \in A+B$. Then, $b+a \in B+A$. Hence, addition in $\mathbf{R}$ is commutative. Now, let there exist an arbitrary $(a+b)+c \in (A+B)+C$. Then, $a+(b+c) \in A+(B+C)$. Hence, addition is associative. □

(c) Show that Definition 8.6.5 (4) holds.

Solution. Let $Y \subseteq Z$,and let $x+y \in X+Y$ with $x \in X$ and $y \in Y$. Then, $y \in Z$, so $x+y \in X+Z$. This implies $X+Y \subseteq X+Z$. □

(d) Show that the cut $O = \{p \in \mathbf{Q} : p < 0\}$ successfully plays the role of the additive identity.

Solution. Let $a+o \in A+O$ where $a \in A$ and $o \in O$. Then, $a+o < a$, implying that $a+o \in A$, so $A+O \subseteq A$. Now, let there an arbitrary $a \in A$ where $a < a' \in A$. Define $s = a+a' < O$, so $A \subseteq A+O$. thus, given $A+O \subseteq O$ and $A \subseteq A+O$, $A = A+O$. □

6 Least Upper Bounds

6.1 Definition 8.6.6:

A set $\mathcal{A} \subseteq \mathbf{R}$ is bounded above if there exists a $B \in \mathbf{R}$ such that $A \leq B$ for all $A \in \mathcal{A}$. The number B is called an *upper bound* for $\mathcal{A}$.

A real number $S \in \mathbf{R}$ is a *least upper bound* for a set $\mathcal{A} \subseteq \mathbf{R}$ if it meets the following two criteria:

1. S is an upper bound for $\mathcal{A}$ and
2. if B is any upper bound for $\mathcal{A}$, then $S \leq B$.

6.2 Exercise 8.6.8

Problem 5. Let $\mathcal{A} \subseteq \mathbf{R}$ be nonempty and bounded above, and let S be the *union* of all $A \in \mathcal{A}$.

(a) First, prove that $S \in \mathbf{R}$ by showing that it is a cut.

(b) Now, show that S is the least upper bound for $\mathcal{A}$

Solution.

(a) Using parts of Definition 8.6.2,

  (i) Since $\mathcal{A} \supset A$, then $\mathcal{A} \neq \emptyset$. Since $\mathcal{A}$ is bounded above by some cut B, $S \leq B$ is implied. Thus, $S \neq \mathbf{Q}$ since S is bounded.

(ii) Let $a \in S$ and $r \in \mathbf{Q}$ with $r < a$. If $a \in S$, then $a \in A$ for some $A \in S$. So, every $r < a$ exists in S.

(iii) Let $a \in S$ where $a \in A$ for some $A \in S$. Therefore, $\exists r \in A$ such that $a < r$ and since $r \in A$, $r \in S$

(b) We know S is an upper bound for $\mathcal{A}$ since $A \subseteq S$ implies $A \leq S$ for all $A \in \mathcal{A}$. Now let B be an upper bound for $\mathcal{A}$. Let $s \in S$ such that $s \in A$ for some $A\imath\mathcal{A}$, implying $a \in S$. Therefore $S \leq B$, making S the least upper bound for $\mathcal{A}$.

□

7 Yay! The Reals are Complete!

This finally proves that $\mathbf{R}$ is complete.

7.1 But wait...Exercise 8.6.9

Problem 6. Consider the collection of so-called "rational" cuts of the form

$$C_r = \{t \in \mathbf{Q} : t < r\}$$

where $r \in \mathbf{Q}$.

(a) Show that $C_r + C_s = C_{r+s}$ for all $r, s \in \mathbf{Q}$. Verify $C_r C_s = C_{rs}$ for the case when $r, s \geq 0$.

Solution. Come back to this. □

(b) Show that $C_r \leq C_s$ if and only if $r \leq s$ in $\mathbf{Q}$.

Solution. Come back to this. □